

Ex:2 TCP CLIENT SOCKET PROGRAMMING IN PYTHON

AIM

To implement and demonstrate client-server communication using socket programming, enabling data exchange over a network.

ALGORITHM

Server Side

1. **Create a socket** using `socket()` function.
2. **Bind** the socket to an IP address and port using `bind()`.
3. **Listen** for incoming connections using `listen()`.
4. **Accept** a connection using `accept()`.
5. **Receive** data from the client using `recv()`.
6. **Send** a response back using `send()`.
7. **Close** the connection using `close()`.

Client Side

1. **Create a socket** using `socket()` function.
2. **Connect** to the server using `connect()`.
3. **Send** data to the server using `send()`.
4. **Receive** response from the server using `recv()`.
5. **Close** the connection using `close()`.

CODE:

Server:

```
import socket
sockfd=socket.socket(socket.AF_INET, socket.SOCK_STREAM)
print('Socket Created')
sockfd.bind(('localhost',55555))
sockfd.listen(3)
print('Waiting for connections')

while True:
    clientfd,addr=sockfd.accept()
```

```
receivedMsg=clientfd.recv(1024).decode()
print("Connected with ",addr)
print("Message Received from Client: ",receivedMsg)
clientfd.send(bytes(receivedMsg,'utf-8'))
print("Message reply sent to Client!")
print("Do you want to continue(type y or n):")
choice=input()
if choice=='n':
    break
```

Client:

```
import socket
clientfd=socket.socket(socket.AF_INET, socket.SOCK_STREAM)
clientfd.connect(('localhost',55555))
name=input("Enter your message:")
clientfd.send(bytes(name,'utf-8'))
print("Message Received from Server: ",clientfd.recv(1024).decode())
```

OUTPUT:

```
C:\Users\Tcs\Desktop>python3 client.py
Enter your message:Hello
Message Received from Server:  Hello

C:\Users\Tcs\Desktop>
```

```
C:\Users\Tcs\Desktop>python3 server.py
Socket Created
Waiting for connections
Connected with ('127.0.0.1', 51204)
Message Received from Client: Hello
Message reply sent to Client!
Do you want to continue (type y or n):
n

C:\Users\Tcs\Desktop>
```

RESULT

The client successfully connected to the server, exchanged messages, and terminated the connection gracefully. This demonstrated the fundamental principles of socket-based communication in a networked environment.